FORECASTING REVENUES

Learning Outcome

- explain approaches to forecasting a company's revenues

After the forecast time horizon is determined, a revenue forecast requires the selection of the forecast objects and approach.

Forecast Objects for Revenues

Forecast objects for revenues are typically either top-down or bottom-up drivers, as discussed in the earlier module on company analysis. Common top-down forecast objects include "growth relative to GDP growth" and "market growth and market share."

Growth relative to GDP growth. The analyst first forecasts the growth rate of nominal GDP. The analyst then considers how the growth rate of the specific company being examined will compare with nominal GDP growth. The analyst can use a forecast for real GDP growth to project volumes and a forecast for inflation to project prices. Analysts often think in terms of percentage point premiums or discounts derived from a company's position in its life cycle (e.g., growth, mature, or decline) or business cycle sensitivity. Thus, an analyst's conclusion might be that a healthcare company's revenue will grow at a rate of 200 bps above the nominal GDP growth rate. The forecast could also be in relative terms. Thus, if GDP is forecast to grow at 4% and the company's revenue is forecast to grow at a 50% faster rate, the forecast percentage change in revenue would be 4% × (1 + 0.50) = 6.0%, or 200 bps higher than the GDP growth rate forecast in absolute terms.

Market growth and market share. The analyst first forecasts a growth rate for a company's product market, and then considers the company's current market share and how that share is likely to change over time. For example, if a company is expected to maintain an 8% market share of a given product market and the product market is forecast to grow from CNY144 billion to CNY154 billion in annual revenue, the forecast growth in company revenue goes from a level of 8% × CNY144 billion = CNY11.5 billion to a level of 8% × CNY154 billion = CNY12.3 billion (considering this product market alone). If the product market revenue has a predictable relationship with GDP, regression analysis might be used to estimate the relationship.

In contrast, an analyst could select bottom-up drivers as the forecast objects, which are then aggregated to arrive at a forecast of total revenue for the company. Examples of bottom-up drivers for revenue forecasts include the following:

- *Volumes and average selling prices*. Forecasts for volumes and prices of the company's products are prepared individually and multiplied to arrive at a revenue forecast. This approach is commonly used for companies that disclose these quantities, such as media and internet companies (users and revenues per user), airlines, asset managers (i.e., AUM and management fee rates), and commodity producers. Many companies disclose price and volume contributions to overall revenue growth, which can be used as the forecast object as well.
- *Product-line or segment revenues*. Forecasts for individual products, product or business lines, geographic areas, or reporting segments are made and then aggregated into a total revenue forecast. This approach is commonly used if a company makes such disclosures and if the objects have different economic exposures.
- *Capacity-based measures.* Forecasts, for example, in retailing, based on the number of stores and sales per store, or same-store sales growth (for stores that have been open

yields on them. For example, net interest income for a bank can be calculated as loans multiplied by the average interest rate minus the product of deposits and liabilities and their average interest rate.

Using elements of both top-down and bottom-up objects can be useful for uncovering implicit assumptions or errors that could arise from using a single approach (e.g., forecasting revenues using bottom-up drivers but checking against what this approach would imply for the company's market size and share).

Separating Recurring and Non-Recurring Revenue or Revenue Growth

An important principle is that non-recurring items and effects should be excluded from a forecast object and considered separately. The non-recurring objects generally do not have the same drivers as the recurring object and may inflate or deflate its size. There are two types of non-recurring items and effects: those that are disclosed by a company's management and those that are not, which require analyst judgment to identify.

For revenue, the first type includes the effect of changes in exchange rates, extra selling days and weeks in the period, acquisitions/divestitures, and other "one-time" revenues or gains. These are separated from a revenue forecast object so that "underlying" revenue or growth can be forecast apart from the non-recurring items (for which a discrete forecast may not be made at all). This is generally a straightforward task unless the credibility of an issuer's financial reporting is in doubt, in which case the analyst may judge certain "non-recurring" items as, in fact, recurring. Some analysts may also use proprietary exchange rate forecasts in revenue projections.

The second type of non-recurring items is those *not* quantified by management and that require analyst judgment to estimate. For example, the COVID-19 pandemic had numerous economic effects, including a rapid increase in e-commerce sales as a percentage of retail sales in most geographies. Many believed that this effect was a "new normal," that the increase in e-commerce sales would recur and subsequent periods would show growth from the newly achieved level. By the second half of 2021, this belief was called into question, because e-commerce sales as a percentage of retail sales began to decline, returning to a trend established prior to the pandemic. Many e-commerce companies saw revenue decelerate or decline in 2022, confirming that some amount of e-commerce's COVID-19-related growth was, in fact, non-recurring. As a second example, graphic processing units (GPUs) are types of semiconductors used in a variety of applications, including gaming PCs. GPU manufacturers saw a large increase in sales in the late 2010s, which many believed was partly attributable to the use of GPUs in cryptocurrency mining, but the companies either could not or would not quantify the precise amount. Although that amount is unknown, an analyst may judge such sales as non-recurring (e.g., due to government restrictions on cryptocurrency mining) and forecast these GPU sales separately from those attributable to gaming.

Forecast Approaches for Revenues

Any of the four forecast approaches introduced in Exhibit 1 in the prior lesson (historical results, historical base rates and convergence, management guidance, discretionary forecasts) can be used for forecasting revenue objects, with the selection based on the analyst's judgment, the information environment, and the issuer's business model. We illustrate how Elaine Nguyen might use the four approaches to forecast revenues for Warehouse Club Inc. in Exhibit 2.

Exhibit 2:

Forecast Approaches for Warehouse Club Inc.'s Revenues

Top-down drivers (e.g., market size, market share)	Historical results	US retail sales excluding autos has grown on average by 3.4% each year for the last 10 years and the company's share of that (70 basis points in 2X19) has increased by 2 basis points per year. Nguyen can forecast revenues by assuming 3.4% market growth and 2 basis points of market share gain each year going forward.
	Historical base rates and convergence	Rather than assuming continued share gains, Nguyen may forecast that Warehouse's sales growth rate will converge to the market growth rate over the next five years, and forecast revenues by assuming a smooth decline in annual market share gains to 0 basis points by 2X25.
	Management guidance	If Warehouse provides annual or longer-term expectations for market size, market share, or sales, Nguyen can use these in her revenue forecast.
	Analyst's discretionary forecast	Warehouse has operated stores in some regional geographies for a long time. Nguyen may commission a study or survey to determine Warehouse's market share in its most established markets (which could be multiples of its all-in market share) and develop market size and share estimates by established/growing/new regions for Warehouse and aggregate them into a sales forecast.

Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
Bottom-up drivers (e.g., stores, sales per store, membership, membership annual price)	Historical results	Warehouse has opened 4–6 new stores, grown net sales per store by 3.8%, and added ~450,000 members per year for the last 10 years. Nguyen could assume these same quantities going forward as the revenue forecast.
	Historical base rates and convergence	There are several mature retailers in the USA that ceased opening new stores several years ago. Nguyen may slow the rate of new-store openings gradually to zero in the USA.
	Management guidance	Management has provided guidance on new-store openings per year and the annual price of membership, which are bottom-up drivers for the revenue forecast.
	Analyst's discretionary forecast	Nguyen may develop a discretionary forecast for stores and members by determining stores per capita and the ratio of members to stores in Warehouse's more mature regional geographies and then applying these quantities, or fractions of them, to different regions where Warehouse is opening stores today. This approach would require a geographic analysis of stores and commissioning surveys to determine regional membership counts or the use of third-party or government data, if available.

Analysts must incorporate their view (implicitly or explicitly) on key risk factors in their revenue forecasts. Four risk factors to consider for all companies are competition, changes in the business cycle, inflation and deflation, and technological developments. For Warehouse Club Inc., two key risk factors are competition and technological developments, embodied in e-commerce, which has been taking share in US retail for many years. Views on these risk factors are also embedded in forecasts of profitability, working capital, capital investments, and capital structure. Given the potential wide range of outcomes, analysts often construct several forecasts for an issuer's financial statements called *scenarios*, each based on different views of key risk factors. We discuss scenario analysis in the last lesson of this module, after forecasting other key financial statement drivers.

Question Set

commerce retailer introduced in earlier modules. Recall that sales by third-party merchants are recognized on a net basis, because Iliso does not take control of the inventory. Iliso receives 15% of the gross merchandise value (GMV) of third-party sales as commission.

	2X19 (USD millions)
National retail sales excluding autos	4,165,000
Iliso's retail sales	7,336
Iliso's third-party merchant sales	1,269
Total GMV	8,605
Iliso's retail sales	7,336
Third-party merchant fees	190
Total Revenues	7,526

Gross merchandise value has increased by an average of 3.6% per year from 2X15 to 2X19 and by 2.7% from 2X18 to 2X19. Iliso's GMV as a percentage of national retail sales has increased steadily, from 0.08% in 2X15 to 0.21% in 2X19.

	2X19	Average Growth per Year, 2X15 to 2X19	Growth from 2X18 to 2X19
Average sales/customer account (USD)	204	5.9%	4.6%
# accounts (millions)	36	22.5%	16.1%
Iliso's retail sales (USD millions)	7,336	29.7%	21.4%
Average sales/third-party merchant (USD)	31,300	13.6%	9.8%
# third-party merchants	40,500	28.5%	28.6%
Third-party merchant sales (USD millions)	1,269	46.0%	41.2%

1. Explain how an analyst would use top-down drivers to forecast Iliso's 2X20 GMV.

Hide Solution

Solution:

An analyst would first forecast national retail sales excluding autos and then forecast Iliso's GMV as a percentage of that market, which represents Iliso's market share. National retail sales excluding autos have been increasing each year, and the analyst is likely to use a growth rate between 2.7% and 3.6%, depending on economic conditions. Iliso's market share in 2X19 was 0.21%, and given past increases, an increase of 2 to 3 bps seems reasonable.

2. Explain why an analyst using top-down drivers needs to allocate Iliso's forecast GMV sales between its own retail sales and sales by third-party merchants. Describe two alternative approaches to doing so.

Hide Solution

Solution:

It is important that an analyst allocate the forecast total GMV between Iliso and third-party merchants, because Iliso's revenue and cost structure differs for each. Iliso recognizes revenue on a gross basis for its own retail sales and on a net basis for sales made by third-party merchants. Iliso's take rate is 15% on third-party merchant sales, but its gross margin percentage is far greater.

forecast increase in GMV sales based on those percentages. Alternatively, an analyst could allocate an increasing proportion to third-party merchant sales, because they have been growing at a faster rate than Iliso's retail sales.

3. Given the following assumptions, the analyst's forecast of Iliso's 2X20 total revenues is *closest* to:

- National retail sales excluding autos grows by 3.4%.
- Iliso's market share (i.e., its GMV as a percentage of national retail sales excluding autos) increases by 2 bps.
- GMV is split 75%/25% between retail sales and third-party merchant sales.
- The take rate on third-party merchant sales remains the same as in 2X19.

A. 7,800.
B. 8,441.
C. 9,906.

Hide Solution

Solution:

A is correct.

Forecast of national retail sales excluding autos = 1.034 × 4,165,000 = 4,306,610

Forecast of Iliso's 2X20 GMV = 4,307,000 × (0.21% + 2 bps) = 9,906

Forecast of Iliso's 2X20 retail sales = 9,905 × 75% = 7,430

Forecast of 2X20 third-party merchant sales = 9,905 × 25% = 2,477

Forecast of 2X20 third-party merchant fees = 2,476 × 15% = 371

Forecast of Iliso's 2X20 total revenues = 7,429 + 371 = 7,801

4. An analyst who uses bottom-up drivers forecasts the number of customer accounts to increase by 15%, the average sales per customer to increase by 5%, the number of third-party merchants to increase by 28%, and the average sales per third-party merchant to increase by 40%. The analyst's forecast for Iliso's 2X20 GMV (in USD millions) is *closest* to:

A. 8,860.
B. 9,210.
C. 11,140.

Hide Solution

Solution:

C is correct.

Forecast of Iliso's 2X20 retail sales = 204(1.05) × 36(1.15) = 8,868 or = 7,336(1.05)(1.15) = 8,858

Forecast of 2X20 third-party merchant sales = [40,500(1.28) × 31,300(1.40)]/1,000,000 = 2,272 or = 1,269(1.28)(1.40) = 2,274

Forecast of Iliso's 2X20 GMV = 8,868 + 2,272 = 11,140

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